

복사전달 특강

Special Topics in Radiative Transfer

Lecture 10
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Chapter 8: MCRT in Outflows and Explosions

Moving media — reference frames, Sobolev approximation, expansion work

- Previous chapters assumed static media or low-velocity environments
Many astrophysical systems involve **large material velocities**: stellar winds, SN ejecta, accretion disk winds
Special relativistic effects — especially the **Doppler effect** — become crucial
- Key challenge: frequency-dependent opacities must be evaluated in the correct frame

8.1 Mixed-Frame Approach

- Two frames: Lab Frame (LF) and Comoving Frame (CMF)
- Packet propagation in LF
- Interactions computed in CMF
- Doppler + aberration transforms

8.2 Line Interactions

- Sobolev approximation
- Optical depth integration simplified
- Re-emission direction sampling
- Test: H Lyman α profile

8.3 Expansion Work

- Energy exchange in moving media
- Packets gain/lose energy in LF
- MCRT naturally captures expansion work

Why Moving Media Require Special Treatment

Breakdown of the static-medium approximation

In **static media**: opacity χ is frequency-independent over the packet trajectory $\rightarrow$ simple integration

In **moving media**: CMF frequency ν_0 changes continuously as the packet propagates through a velocity gradient

- The optical depth integral must account for the Doppler-varying opacity along the path

$$\tau(l) = \int_0^l dl' \chi(l') = \int_{\nu_0(0)}^{\nu_0(l)} d\nu_0 \frac{dl}{d\nu_0} \frac{\nu_0}{\nu} \chi_0(\nu_0) \quad [\text{Eq. 79}]. \quad \Leftarrow \quad \nu \chi_\nu = \text{Lorentz invariant, and } \tau_\nu = \text{Lorentz invariant.}$$

The factor ν/ν_0 transforms the opacity from CMF to LF; $d\nu_0/dl$ encodes the Doppler shift along the path.

General Case: Computationally Expensive

- Must integrate frequency change along trajectory
- Opacity varies strongly near line resonances
- Requires re-evaluation at each propagation step

Sobolev Approximation

- Velocity field monotonically varying $\rightarrow$ each line resonance encountered once
- Reduces to a purely local problem at the Sobolev point
- Enables dramatic computational speedup

8.1 The Mixed-Frame Approach

Lab Frame (LF) propagation + Comoving Frame (CMF)

Lab Frame (LF): defined so the computational grid is at rest — natural for tracking positions and times

Comoving Frame (CMF): local rest frame of the material — radiation-matter interaction physics takes simplest form

- Mixed-frame approach: propagate packets in LF, but treat all interactions in CMF

****Key idea**:** the handling of different tasks in MCRT simulations is easier in one or the other frame.

→ Spatial/temporal grid: defined in LF ; Opacity/emissivity: evaluated in CMF

We assume a simplified situation of coherent and isotropic scattering. These assumptions will only ever hold in CMF.

Whenever a MC packet interacts, its properties are transformed into the CMF, where the interaction is simulated.

Doppler Shift (LF → CMF)

incident CMF frequency: $\nu_{0,i} = \gamma \nu_i (1 - \boldsymbol{\beta} \cdot \mathbf{n}_i)$

- $\boldsymbol{\beta} = \mathbf{v}/c$ (material velocity)
- $\gamma = (1 - \beta^2)^{-1/2}$
- $\nu_{0,i}$: incident CMF frequency
- $\mathbf{n}_i$: incident propagation direction (LF)

Energy Transform (LF → CMF)

$\varepsilon_{0,i} = \gamma \varepsilon_i (1 - \boldsymbol{\beta} \cdot \mathbf{n}_i)$

- Coherent scattering in CMF: $\varepsilon_{0,e} = \varepsilon_{0,i}$ and $\nu_{0,e} = \nu_{0,i}$
- Re-transform into LF after interaction.

$\varepsilon_e = \gamma \varepsilon_{0,e} (1 + \boldsymbol{\beta} \cdot \mathbf{n}_{0,e})$

$\nu_e = \gamma \nu_{0,e} (1 + \boldsymbol{\beta} \cdot \mathbf{n}_{0,e})$

0 = quantities measured in CMF.

i, e = incident and emergent packet properties

8.1 Aberration and Packet Initialization

Direction transforms and CMF initialization of packets

Aberration: propagation direction transforms between LF and CMF (in addition to Doppler)

- In many applications, the frequency shifts due to the Doppler effect will be the most significant consequence of the material motion.
- However, for a detailed relativistic treatment, aberration effects must be taken into account when converting incident direction to CMF and emergent direction back to LF

Incident Direction Transform: LF $\rightarrow$ CMF

$$\mathbf{n}_{0,i} = \left(\frac{\nu}{\nu_0} \right) \left[\mathbf{n}_i - \gamma\beta \left(1 - \frac{\gamma}{\gamma+1} \mathbf{n}_i \cdot \boldsymbol{\beta} \right) \right]$$

- Used to convert direction before interaction
- Necessary for achieving at least $O(v/c)$ accuracy.

Emergent Direction Transform: CMF $\rightarrow$ LF

$$\mathbf{n}_e = \left(\frac{\nu}{\nu_0} \right) \left[\mathbf{n}_{0,e} + \gamma\beta \left(1 + \frac{\gamma}{\gamma+1} \mathbf{n}_{0,e} \cdot \boldsymbol{\beta} \right) \right]$$

- Applied after interaction in CMF
- Re-emission direction set in CMF first

Optical Depth Accumulation (Moving Media)

- Care has to be taken when calculating the accumulated optical depth τ . The integration is best performed in LF, since the propagation is carried out in LF.

$$\tau(l) = \int_{\nu_0(0)}^{\nu_0(l)} d\nu_0 \frac{dl}{d\nu_0} \frac{\nu_0}{\nu} \chi_0(\nu_0)$$

- Here, the line opacities is transformed : $\chi_{\text{LF}} = \chi_{\text{CMF}} \cdot (\nu_0/\nu)$.
The CMF frequency changes continuously along path by $dl/d\nu_0$.
- In general, performing this integration is very challenging since it has to be carried out “frequently” during the propagation of each packet.

CMF Packet Initialization

- One also has to decide in which frame the MC packets are launched during the initialization.
- Often, CMF is the natural choice, e.g., thermal emission sampled from Planck in CMF
- In such cases, packet properties are drawn in CMF and then transform into LF.
- Ensures correct energy conservation from the start

8.2 Line Interactions in Outflows: Sobolev Approximation

Simplifying bound-bound opacities in monotonically expanding flows

- Treating frequency-dependent opacities is challenging. However, in the case of bound-bound (line) processes, which are **resonant**, the situation can be significantly simplified in the presence of large material velocities.
- Only photons with CMF frequency $\approx \nu_{lu}$ interact. In a monotonically expanding flow, photons redshift continuously — each line resonance is encountered **once**.
- Over the line width (mainly determined by thermal + turbulent motion), the plasma state is approximately constant $\rightarrow$ Sobolev approximation.
- **Sobolev length:** over which the photon frequency shifts into and out of resonance with the line in CMF.

This scale is compared with the typical length over which the plasma state, and thus frequency-independent parts of the line opacity change. If the Sobolev length is much smaller, the line profile can be approximated by a delta-distribution around ν_{lu} .

Sobolev approximation: replace the full frequency-dependent line profile by a delta-function at ν_{lu} .

$\rightarrow$ Optical depth integration collapses to a single local evaluation at the Sobolev point r_s .

This approximation reduces the calculation of the line optical depth to a **local problem**.

In addition, line overlaps are eliminated. Since photons continuously red shift in monotonously expanding flow (or blue shift in compression flows), **they successively scan over the possible line transitions in the CMF one-by-one**.

This suggests storing **the line transitions in a frequency-ordered list**.

Sobolev Optical Depth

Whenever the packet reaches the Sobolev point of the next line transition, i.e., when its local CMF frequency equals the natural frequency of the line, the accumulated optical depth is instantaneously incremented by the full line opacity of the transition:

$$\tau_s = \frac{\pi e^2}{m_e c} f_{lu} n_l \left(1 - \frac{n_u}{n_l} \frac{g_l}{g_u} \right)_{r_s} \left(\frac{dl}{d\nu_0} \right)_{r_s}$$

- f_{lu} : absorption oscillator strength $l \rightarrow u$
 n_l, n_u : level populations
 g_l, g_u : statistical weights
 $(dl/d\nu_0)$: velocity gradient term
- Here, r_s denotes that the calculation is evaluated locally at the Sobolev point.

Homologous Flow Simplification

In one-dimensional geometries, the velocity gradient term simplifies to

$$\left(\frac{dl}{d\nu_0} \right)_{r_s} = \frac{c}{\nu_{lu}} \left[\frac{1}{(1 - \mu^2)v/r + \mu^2 dv/dr} \right]_{r_s}$$

- In homologous flow, $v/r = \text{const.}$
- Escape probability becomes direction-independent
- Greatly simplifies re-emission direction sampling

8.2 MCRT Implementation of the Sobolev Approximation

Packet propagation scheme with Sobolev line interactions

- MC packets accumulate **continuum** optical depth τ_{cont} as they propagate (frequency-independent part).
- Simultaneously, track distance to next **Sobolev point** for each line in a frequency-ordered list.
- If the optical depth value is reached at a Sobolev point, the packet performs the corresponding interaction.
- Decision at each step: does the packet reach a Sobolev point before τ_{cont} is exhausted?

Continuum Interaction (τ_{cont} reached first)

- Packet moved to distance $l \Leftarrow \tau_p(l) = \int_0^l dl' \chi(l')$.
- Interaction type sampled from continuum processes
- Sobolev line contributions reset/recalculated
- e.g., Thomson scattering, photoionization

Line Interaction (Sobolev point reached)

- Packet moved to Sobolev point r_s , and τ_{cont} instantaneously incremented by full τ_s
- Sample: scatter (resonant) or pass through
- It should be noted that the re-emission direction should be drawn from escape probability:

$$\rho(\mathbf{n}) \propto \frac{1 - \exp(-\tau_s(\mathbf{n}))}{\tau_s(\mathbf{n})}$$

The Sobolev approach stores lines in a frequency-ordered list (Abbott & Lucy 1985; Mazzali & Lucy 1993). As the packet redshifts monotonically, it sweeps through lines one by one — a hugely efficient scheme.

8.2 Sobolev Optical Depth Accumulation

Fig. 7 — Illustration of the optical depth accumulation process

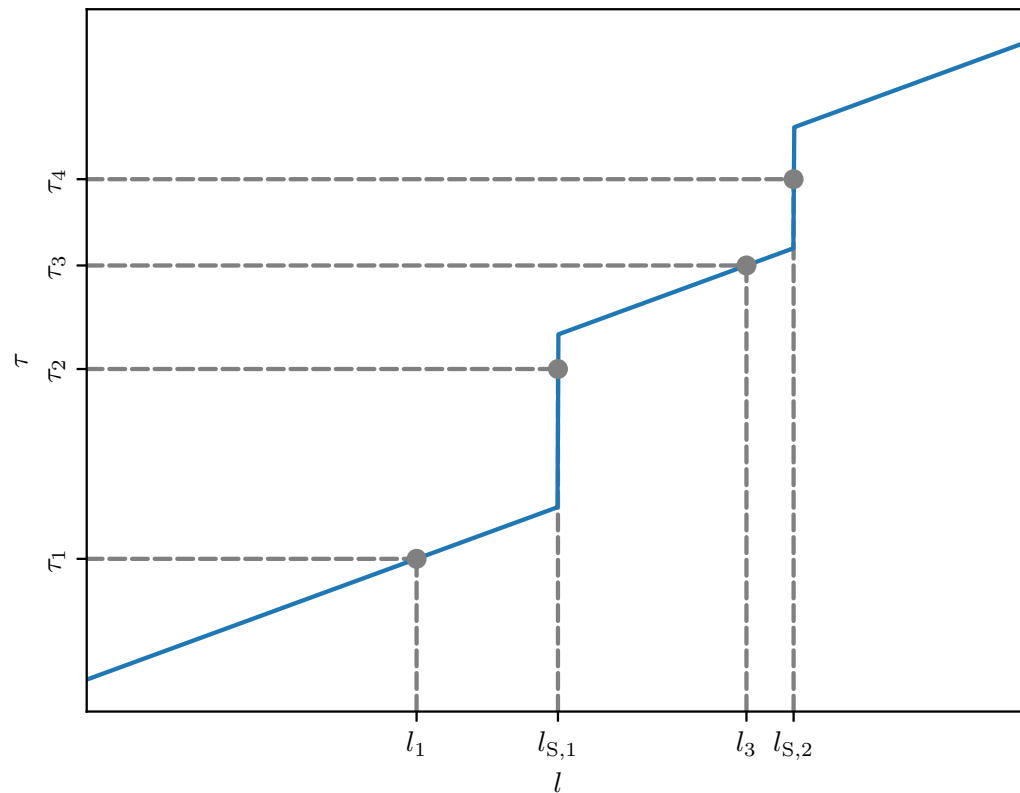


Fig. 7: Optical depth τ accumulates continuously from continuum processes. At two Sobolev points S_1 and S_2 , τ is instantaneously incremented by the full Sobolev line depth. For τ_1 and τ_3 : packet undergoes a continuum interaction at l_1 or l_3 . For τ_2 and τ_4 : packet undergoes a line interaction at the corresponding Sobolev point.

8.2 Test: H Lyman α Line Profile in a Homologous Flow

Fig. 8 — Validation of the Sobolev-based MCRT scheme

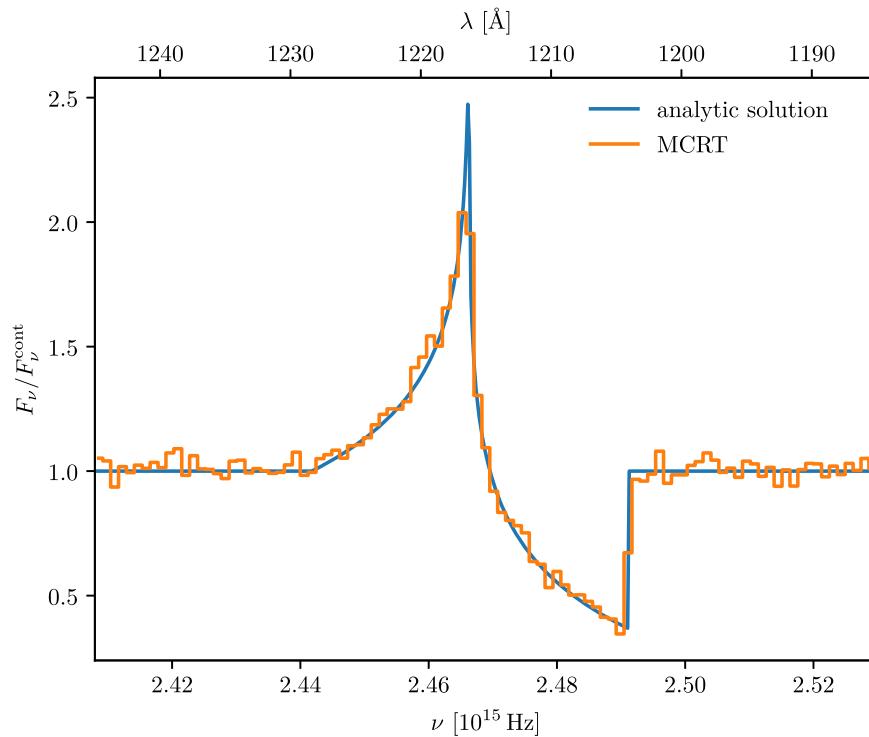


Fig. 8: H Lyman α line profile from a simple mixed-frame MCRT simulation. Emergent spectra normalized by the incident continuum flux F_ν^{cont} .

Blue: analytic solution (Jeffery & Branch 1990).
Orange: MCRT result.

Excellent agreement validates the mixed-frame + Sobolev MCRT approach.

Test Setup (Appendix A.2)

- H sphere, $\tau_s = 1$ for Ly α , $N = 10^5$ packets
- Analytic: Jeffery & Branch (1990)

<https://www.physics.unlv.edu/~jeffery/astro/radtran/>

Result

- MCRT matches analytic solution closely
- MC noise visible but statistically valid

8.3 MCRT and Expansion Work

Energy exchange in mass outflows: the radiation field drives the expansion

- In RE, packet energy is conserved in the **CMF** during interactions.
But in the **LF**, energy is NOT conserved: packets can gain or lose energy at each scattering.
- When a packet interacts, the mixed-frame MCRT approach switches to the local CMF to perform the interaction and update the packet properties before returning into the LF.
For instance, in isotropic and resonance scatterings, the LF energy of a packet changes during the interaction from ε_i to ε_e according to

$$\varepsilon_e = \varepsilon_i \frac{1 - \boldsymbol{\beta} \cdot \mathbf{n}_i}{1 - \boldsymbol{\beta} \cdot \mathbf{n}_e}$$

Depending on the orientation of the propagation direction prior and after the scattering relative to the material flow, the packet gains or loses energy in the LF.

- Photons propagating **outward** in an outflow lose energy on average → this is **expansion work**.

In astrophysical mass outflows, radiation is typically emitted from a central source (a star or an accretion disk). Then, photons predominantly propagate in the direction of the flow before they interact.

Electron scatterings or resonant line interactions are either approximately isotropic or forwards-backwards symmetric. ⇒ A radiation field that is initially aligned with the expanding flow will be partly diffused by the interactions and packets (on average) lose energy in the LF.

Energy exchange in mass outflows: the radiation field drives the expansion

Mean LF energy after isotropic re-emission from a packet with incident direction μ_i will be given by

$$\bar{\epsilon}_e = \frac{1}{2} \epsilon_i \int_{-1}^1 \frac{1 - \mu_i \beta}{1 - \mu_e \beta} d\mu_e \quad (\text{assumed spherical symmetry and have neglected aberration}).$$

Performing the integration gives:

$$\frac{\bar{\epsilon}_e}{\epsilon_i} = \frac{1}{2\beta} (1 - \mu_i \beta) [\log(1 + \beta) - \log(1 - \beta)] \rightarrow \frac{\bar{\epsilon}_e}{\epsilon_i} = 1 - \mu_i \beta + \mathcal{O}(\beta^2) \text{ for small } \beta$$

For $\mu_i = 1$ (outward propagating): $\bar{\epsilon}_e < \epsilon_i$ — net energy loss $\rightarrow$ expansion work!

Physical Interpretation

- Radiation field initially aligned with outflow.
- After multiple scatterings, packets lose energy in LF. Energy transferred to kinetic energy of the outflowing gas.
- This is the radiative driving mechanism in hot-star winds.

MCRT Naturally Captures This

- No special treatment needed in mixed-frame MCRT
- Expansion work = $\sum (\epsilon_i - \epsilon_e)$ over all interactions
Tracked by comparing incident and emergent LF energies.
- Lucy (2005): model SN test demonstrates this directly

8.3 Test: Lucy (2005) Model Supernova

Figs. 9 & 10 — Energy flows and expansion work in a simplified SN ejecta model

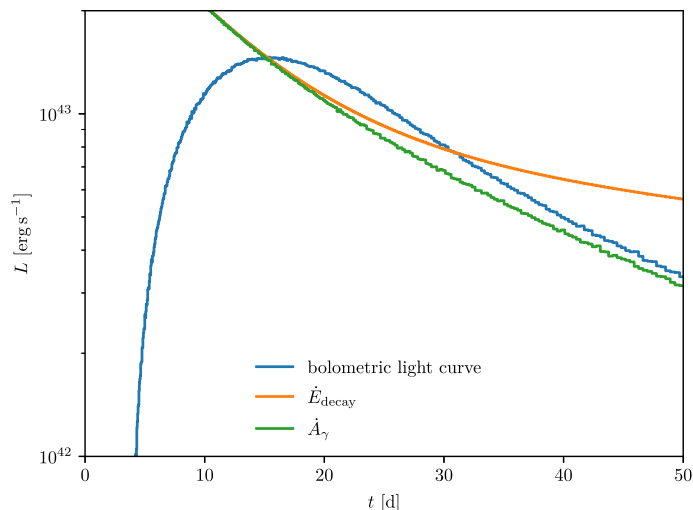


Fig. 9: Bolometric light curve from the MCRH code for the Lucy (2005) model SN.

Blue: escaping luminosity.

Orange: energy release rate from ^{56}Ni radioactive decay chain.

Green: rate at which γ -packets deposit energy into the UV-optical-IR radiation field.

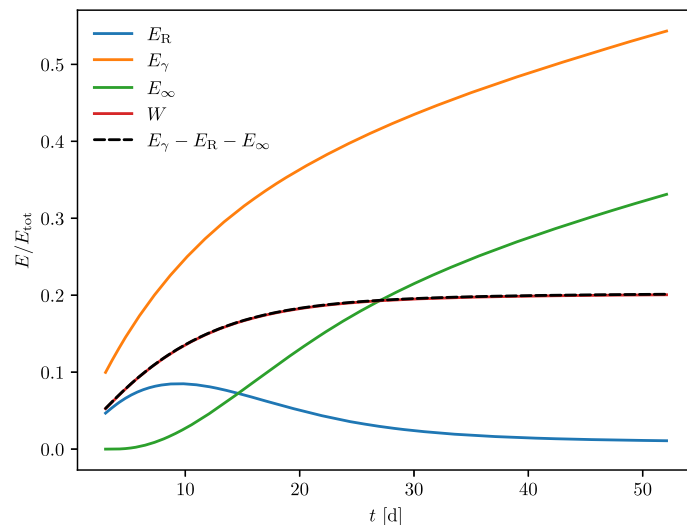


Fig. 10: Energy flows — $E_R(t)$ (energy stored in the radiation field), $E_\infty(t)$ (energy that has escaped to infinity), $E_\gamma(t)$ (energy released by radioactive decays), $W(t)$ (work term obtained directly from the MCRT simulation by balancing the emergent and incident LF packet energies in the interaction). The black dashed curve $E_\gamma - E_R - E_\infty$ precisely follows W , confirming energy conservation.

Key result: Mixed-frame MCRT correctly captures expansion work without special treatment.

Energy conservation verified at every time step across the simulation.

Chapter 8 — Summary

MCRT in outflows: mixed-frame, Sobolev, expansion work

Mixed-Frame Approach

- LF propagation + CMF interactions
- Doppler (Eqs. 71–76) + aberration (Eqs. 77–78)
- Optical depth accounts for CMF frequency drift
- Packet initialization often in CMF

Sobolev Approximation

- Monotonic velocity field → each line once
- τ_s : purely local evaluation at Sobolev point
- Line interactions: sampled via escape probability
- H Lyman α test: excellent agreement

Expansion Work

- LF energy NOT conserved per interaction
- Net energy loss = expansion work done on gas
- MCRT captures this automatically
- Validated with Lucy (2005) model SN

- These techniques are widely used: Abbott & Lucy (1985), Mazzali & Lucy (1993), Vink et al. (1999, 2000), Sim (2004), Noebauer & Sim (2015)...

MCRT + Sobolev is **the** standard approach for spectral synthesis in SN ejecta and hot-star winds

Next: how to **extract** physical information from MCRT simulations more efficiently (Chapter 9)